Denoising Diffusion Models: A Short Primer

Abstract. Diffusion models generate images by learning to reverse a gradual noising process. A forward process adds a small amount of Gaussian noise to an image over many steps until only noise remains. A neural network is trained to predict the noise that was added at each step, and sampling runs the chain backwards: it starts from pure noise and removes a little of it at a time.

This note explains the forward process, the training objective and the sampling loop, and compares diffusion models with generative adversarial networks and variational autoencoders.

1. The forward process

At step t the image x_t is a weighted mix of the previous image and fresh Gaussian noise. The noise schedule beta_1 ... beta_T controls how quickly information is destroyed. Because a sum of Gaussians is Gaussian, x_t can be sampled from the clean image x_0 in a single step.

2. Training

The network sees a noisy image and the step number, and predicts the noise. The loss is the mean squared error between the true noise and the predicted noise, averaged over random steps.

3. Sampling
Start from x_T drawn from a standard normal distribution. For t = T down to 1, subtract the predicted noise, rescale, and add a small amount of fresh noise. After T steps the result is a sample from the learned distribution.

4. Trade-offs
Sampling needs hundreds of network evaluations, so it is slower than a GAN, but training is stable and the samples cover the data well. Later work reduces the number of steps with better samplers and distillation.